Comparison of Lightweight Encryption Algorithms

B.Tech Final Year Project

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Outline

Motivation and Problem Statement

Context

- IoT deployments require cryptography on devices with limited CPU, memory, and energy
- Conventional algorithms (AES, RSA, ECC) often exceed microcontroller resource limits
- The Mirai botnet attack highlighted risks of insecure embedded systems

Problem

- Developers sometimes deploy weakened or ad hoc security mechanisms
- Lightweight cryptography provides efficient alternatives for constrained platforms
- Need to balance security, performance, and implementation cost

Objectives & Contributions

This study aims to:

- Evaluate PRESENT, SPECK, and ASCON on a Raspberry Pi Pico (ARM Cortex-M0+)
- Compare execution time, memory usage, and throughput
- Identify design trade-offs for constrained embedded environments

Contributions:

- Reproducible Rust-based benchmarking framework for the RP2040
- Empirical data across block cipher, mode, and payload-size dimensions
- Practical deployment recommendations based on measured trade-offs

Design Properties Comparison

Property	**PRESENT**	**SPECK**	**ASCON**
Block Size	64 bits	64 bits	64 bits
Key Sizes	80, 128 bits	64–256 bits	128 bits
Structure	SPN	ARX Feistel	Sponge-Duplex
Rounds	31/32	22-34	12
Year	2007	2013	2014
Standardization	ISO/IEC 29192-2	—	NIST SP 800-232
AEAD Support	No	No	Yes

Table: Design Properties Comparison of Evaluated Ciphers

PRESENT

Key Properties:

- 64-bit SPN block cipher, 80/128-bit keys
- 1,570 GE gate count – among the most compact ciphers
- Round steps: AddRoundKey, 16 $\times$ 4-bit S-box layer, bit permutation
- 31 rounds (PRESENT-80), 32 rounds (PRESENT-128)
- Standardized in ISO/IEC 29192-2

SPECK

Key Properties:

- ARX-based Feistel-like structure (add, rotate, XOR only)
- Highly efficient on microcontrollers
- Word sizes: 16, 24, 32, 48, 64 bits
- SPECK-64/128: 27 rounds, $\alpha = 7, \beta = 2$ rotation constants
- Fastest lightweight cipher in software benchmarks

ASCON

Key Properties:

- NIST Lightweight Cryptography Standard (SP 800-232, 2023)
- Sponge-duplex construction, 320-bit internal state
- Built-in authenticated encryption and hashing (AEAD)
- Two permutation variants: p^a (12 rounds) and p^b (6 rounds)
- 64-bit rate / 256-bit capacity – provides 128-bit security

Platform & Evaluation Metrics

Hardware: Raspberry Pi Pico (ARM Cortex-M0+, 133 MHz, 264 KB SRAM, 2 MB Flash)

Software: Rust `no_std`, `rp2040-hal`, release mode with LTO

Metrics:

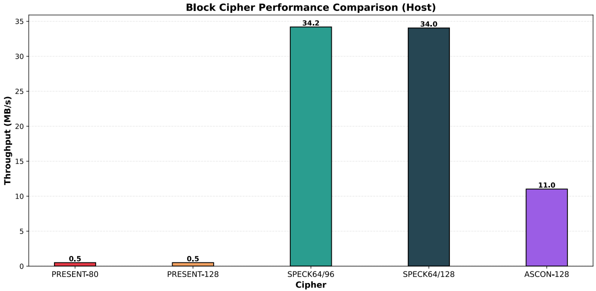
- **Execution Time**: hardware timer, interrupts disabled
- **Cycles Per Byte**: $CPB = \frac{\text{total cycles}}{\text{total bytes}}$
- **Throughput**: Mbps from fixed-size buffer encryption
- **Memory**: ROM (binary size) + RAM (state/buffer analysis)

Procedure: 1000+ iterations, warm-up runs, cross-validated with test vectors

Block Cipher Benchmarks (Raspberry Pi Pico)

Algorithm	CPB	Mbps
PRESENT-80	2,958	0.34
PRESENT-128	3,068	0.33
SPECK-64/96	42	23.81
SPECK-64/128	42	23.67
ASCON-128	132	7.56

Key Finding: SPECK is **70x faster** than PRESENT and **3x faster** than ASCON.



Memory Footprint

Cipher	Key	State	ROM	RAM
PRESENT-80	80 bits	8 bytes	$\sim$2 KB	$\sim$16 B
PRESENT-128	128 bits	8 bytes	$\sim$2.5 KB	$\sim$16 B
SPECK-64/96	96 bits	16 bytes	$\sim$1 KB	$\sim$32 B
SPECK-64/128	128 bits	16 bytes	$\sim$1.2 KB	$\sim$32 B
ASCON-128	128 bits	40 bytes	$\sim$3 KB	$\sim$40 B

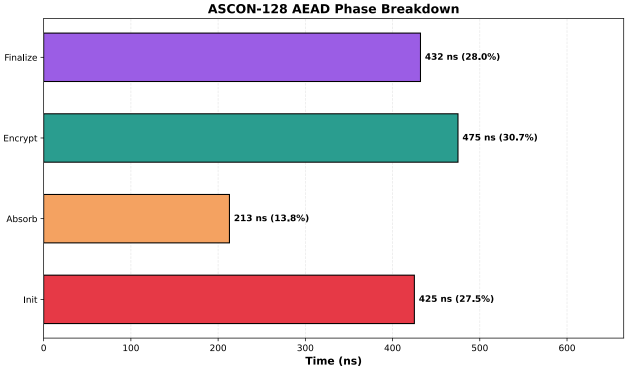
Table: Memory Footprint Comparison

Key Finding: SPECK has the smallest ROM footprint ($\sim$1 KB) due to ARX-only operations (no lookup tables).

ASCON Phase Breakdown

Phase	Cycles
Initialization	427
Absorb (AD)	216
Encrypt	464
Finalize	422
Total	**1,529**

Table: ASCON AEAD Phase Breakdown

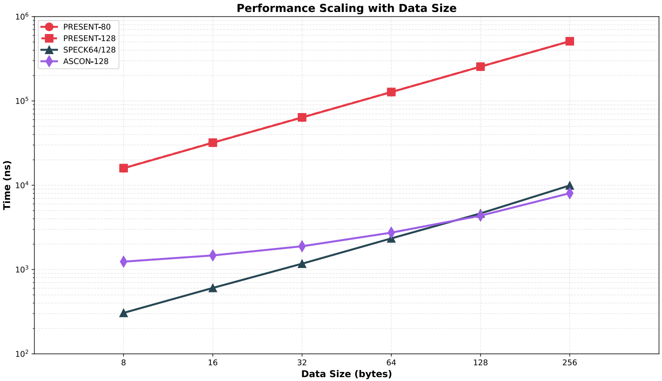


Speedup & Scaling Analysis

Performance relative to ASCON:

Cipher	64B	256B	1
SPECK (ECB)	1.34x	1.08x	:
PRESENT (ECB)	0.012x	0.010x	0

Key Insight: SPECK's edge narrows as ASCON's 427-cycle init overhead amortizes over larger payloads.



Key Takeaways

- **SPECK**: Best raw throughput (23.67 Mbps) – optimal for bulk encryption on microcontrollers
- **ASCON**: Competitive throughput (7.56 Mbps) with built-in AEAD – gap narrows to $\sim$10% at 256B+ payloads
- **PRESENT**: 70x slower in software but remains relevant for hardware-constrained RFID-class devices
- **Mode selection matters**: CTR offers semantic security with only 11% throughput overhead vs ECB

Scorecard Comparison

Metric	**PRESENT**	**SPECK**	**ASCON**
Throughput (Mbps)	0.34	23.67	7.56
CPB (cycles)	2,958	42	132
ROM (KB)	2.5	1.2	3.0
RAM (bytes)	16	32	40
Authenticated Encryption	No	No	Yes
NIST Standard	No	No	Yes (SP 800-232)
ISO Standard	Yes (29192-2)	No	No
Hardware Efficiency	Excellent	Good	Good
Software Efficiency	Poor	Excellent	Good

Table: Scorecard Comparison: PRESENT, SPECK, ASCON

Use-Case Recommendations

1. **Real-time sensor data encryption**: SPECK-64/128 in CTR mode
    - 23.67 Mbps, 32 bytes RAM, ideal for high-bandwidth sensor streams
2. **Secure IoT communication with authentication**: ASCON
    - 7.56 Mbps, built-in AEAD, NIST standardized (SP 800-232)
3. **Hardware-constrained RFID tags**: PRESENT
    - Ultra-low gate count, 0.34 Mbps adequate for low-bandwidth communications

Thank You

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